Agentic MT: Initial Experiments

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Six phases testing the two-pillar theory (slow thinking + tool use, jointly necessary, most valuable for low-resource pairs). Code and full per-phase logs: `github.com/imthomp/agentic-mt`.

Where the theory stands

The central joint-necessity claim has not been confirmed or falsified — every direct test has hit a model confound first. We do have real evidence against the simplest instantiations of two supporting mechanisms (QE-in-the-loop, source-side triggering), and a new finding, from one model across five language points (BLOOMZ, Phase 6), that raw pretraining exposure to a language can be a hard gate on generation — more a threshold effect than a smooth curve — separate from anything the agentic scaffolding does.

Phase-by-phase

Phase	Setup	Finding
1: QE reliability	COMET-KIWI vs. real WMT DA human scores, 5 low-resource pairs (ha/km/ps/xh/zu), 1000-sample bootstrap	Below 0.3 for 3/5 pairs (ha 0.115, ps 0.295, zu 0.247); km/xh weak but usable
2: Triggering	5 source-side features + logistic regression on Phase 1's data, same 5 pairs ($n = 38786$)	No feature or model beats chance (AUC 0.513–0.564); pooled OOV signal ($r = -0.33$) is a Simpson's-paradox artifact
3: Joint necessity	Aya-101, 4 conditions $\times$ en-de/en-ha $\times$ 100 sentences	CoT chain degenerates into verbatim repeats in most sentences (Fig. 1), confounding the comparison; raw scores show tool-alone highest and CoT lowest ($p = 0.004$–0.024 for the resource-level gap), but this likely reflects the degeneracy, not reasoning quality
4: Refinement	Qwen2.5-32B, TEaR self-critique vs. external-QE refine, en-de, 100 sentences, 4 iterations	Neither condition peaks-then-declines; both improve gradually to iteration 4, matching TEaR's actual reported pattern
5: Model search	Tried BLOOMZ, InkubaLM as open, conversation-tuned models covering ha/km/ps/xh/zu	Both ruled out: InkubaLM incoherent even in English (0.4 B params); BLOOMZ needed 9 fixes, then failed on exposure grounds
6: Exposure floor	BLOOMZ smoke test, en-fr/en-sw/en-yo/en-xh/en-zu, 10 sentences/pair, hand-categorized	Clean-generation rate never rises as pretraining bytes fall, but drops sharply rather than smoothly, this model only (see Fig. 2)

Findings, by implication

None of the thresholds/p-values below correct for multiple comparisons across the six phases — read each as suggestive on its own terms, not as part of a jointly-controlled family.

Challenges to specific proposed mechanisms

- **QE-in-the-loop (Sec. 5.3) is not viable as specified for most of the 5 low-resource pairs tested (ha/km/ps/xh/zu).** An en-de benchmark of ≈ 0.57 sets the bar these pairs miss. This confirms, rather than merely illustrates, the risk the proposal itself flags in Sec. 5.3 ("reference-free QE models... must be adapted for language pairs with limited QE training data") —

the mechanism's foundation is measurably weaker exactly where the theory says it should matter most. To trust more: extend beyond the five pairs with real human judgments; current coverage is a hard ceiling on what we can check.

- **Source-side triggering (Sec. 5.1) shows no signal from three of the five features that section names** (OOV rate, sentence length, idiomatic density) **on these same 5 pairs,** against a chance baseline of 0.5. The pooled OOV correlation evaporates to -0.045–0.021 once controlled for language. To trust more: a real domain classifier and a frequency-list OOV measure (the two named features not yet tested) before calling triggering a dead end rather than an implementation gap.

The central joint-necessity claim: confounded, not resolved

- **Phase 3**: Aya-101 is not conversation-tuned, and its CoT chain produced byte-identical draft/refine/proofread output in 80–86 of 100 sentences (Fig. 1) — this degeneracy confounds any reasoning-quality comparison before we even look at scores. Given that: the raw numbers (tool-alone wins both pairs; CoT scores lowest; gap differs by resource level, $p = 0.004$–0.024) describe what happened, but should be read as "a broken CoT chain scored worse," not as evidence about reasoning per se. Worth flagging even with that discount: CoT+Tool scored below Tool-alone on *both* pairs — the joint-necessity claim predicts the combination should beat either pillar alone, and the raw direction here runs the other way. The degeneracy is a sufficient explanation, but we haven't shown it's the *only* one; this isn't evidence for the theory even discounted.
- **Phase 4**: matches Feng et al.'s actual TEaR pattern, not the early-peak hypothesis we set out to test (that pattern belongs to a different paper). Secondary finding: condition A froze only 31/100 sentences by iteration 4 vs. condition B's 48/100 by iteration 3, and A's trajectory is smoother — span-level feedback may matter more than feedback source. To trust more: rerun condition B with XCOMET (span-level) to isolate granularity from source. *Relevance to the central claim:* this phase tests refinement quality, not joint necessity directly — it neither confirms nor challenges the central claim, though the granularity finding is worth folding into whichever joint-necessity test runs next. *Gap against the proposal:* Sec. 5.3 predicts the undirected-vs-QE-directed refinement gap "should be largest in low-resource settings"; this phase tested exactly those two conditions but only on en-de, so it cannot speak to that specific prediction — still open.
- **Phases 5–6**: two replacement-model candidates ruled out with direct evidence, not assumption (see table). **Net: the joint-necessity question is still untested on a model without a disqualifying confound.**
- **New: best-of-5 Level-1 baseline, filling Sec. 7's (b)-vs-(c) gap.** Sec. 7 specifies comparing inference-time-scaled Level 1 (best-of-N/MBR) against full agentic Level 2 as "particularly important"; none of Phase 3's four conditions did this. Ran it for en-ha: best-of-5 temperature sampling, no tools, no reasoning, scored 0.666 COMET-KIWI — higher than *all four* architected conditions, including tool-alone (0.602). Taken at face value this is an uncomfortable answer to Sec. 7's question: compute alone beat architecture. **But it can't be taken at face value:** selection here used COMET-KIWI as judge, and Phase 1 measured COMET-KIWI's correlation with real human judgment for Hausa specifically at just 0.115 — the weakest of all five pairs. This may be rewarding samples that exploit COMET-KIWI's blind spots rather than ones that are genuinely better. **Tried to rerun on a QE-trustworthy low-resource pair (km, $r = 0.457$) and found there isn't one:** that correlation is measured km→en, not en→km — Phase 1's data has no en-km rows at all, and the same holds for ps (ps-en only) and xh/zu (xh-zu/zu-xh only, never against English). Hausa is the *only* tested pair with real en→X human-

judgment data, and it's the one already flagged as unreliable — there is currently no low-resource fallback. **Reran instead on en-de** (directionally valid QE reference ≈ 0.57, existing four-condition baseline): best-of-5 scored 0.827, again edging out tool-alone (0.819) — but by 0.007, against en-ha's 0.064 gap. With a more trustworthy judge, "compute alone wins" nearly vanishes to a rounding-distance tie, consistent with (not proof of) the en-ha margin being substantially QE noise rather than a real compute-beats-architecture result. Still doesn't settle Sec. 7 for the low-resource case en-de doesn't represent, but it changes how much weight the en-ha number alone deserves.

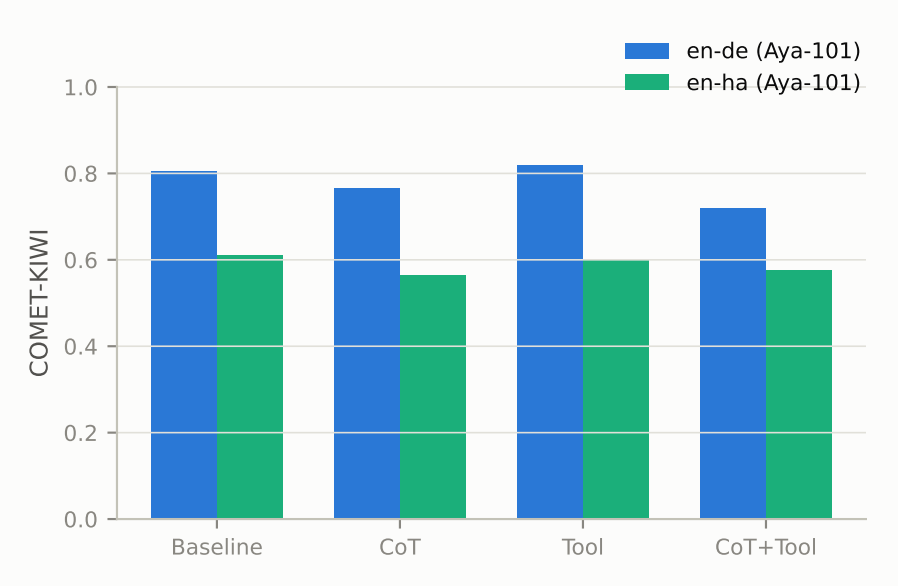


Figure 1: Phase 3 condition means (COMET-KIWI). CoT's low score reflects the degenerate-repeat confound described above, not a clean reasoning-vs-baseline comparison.

Tangential but new: a pretraining-exposure floor

- **Phase 6**: below some exposure level BLOOMZ doesn't attempt the target language at all — it defaults to echoing the English source (comprehension intact, generation not) — rather than degrading gracefully. Adding Yoruba (∼90M ROOTS bytes, between Swahili's 236M and Xhosa/Zulu's 14.3M/8.5M) as the planned intermediate point produced 0% clean output, matching Xhosa/Zulu rather than sitting between them and Swahili's 50% — so this looks like a threshold somewhere between 90M and 236M bytes, not the smooth dose-response curve the four-point version suggested. To trust more: still only 5 points (this model only); a language between 90M and 236M bytes would tell us whether this is a sharp cutoff or just a steep slope. *Challenge to Sec. 4, sharpened:* Sec. 4's "more marginal value as coverage thins" claim is plausible in the middle regime (model attempts translation but lacks vocabulary/nuance) but breaks at the floor for three compounding reasons, not just one. *Conceptual:* below the floor there's no translation quality for a resource to have marginal value *over* — source-echo isn't a bad translation, it's a non-translation. *Mechanistic:* external resources are usually delivered *in* the target language; if the model's representation of that language is too sparse to translate, it may be too sparse to use the resource either — help arrives through the same broken channel. *Policy:* the comparison this claim is meant to inform reverses below the floor — pretraining-time intervention beats inference-time scaffolding once the problem is missing capability, not missing knowledge. The bounded, defensible version of Sec. 4's claim ("above the floor, marginal value increases as coverage thins") is close to tautological — it says knowledge-gap fixes help

knowledge gaps. The consequential version — that this extends to the lowest-coverage languages, which is what motivates prioritizing them — is exactly what the floor finding argues against.

- Cross-checked for Phase 3's Aya-101/mT5: German has 347B tokens in mC4 vs. Hausa's 0.2B ($\sim 1735\times$) — so Phase 3's en-de anchor *is* legitimate for that model, unlike BLOOM's zero-exposure German. Same question, opposite answer for the two models — check per-model, don't assume.

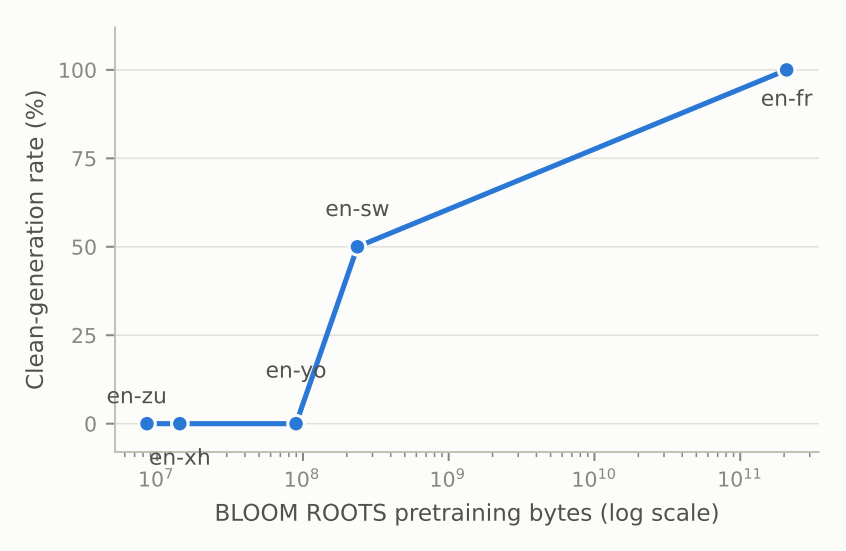


Figure 2: BLOOMZ clean-generation rate vs. ROOTS pretraining bytes (log scale), en-fr/en-sw/en-yo/en-xh/en-zu. Note the drop between en-sw and en-yo, not a smooth ramp.

What we know regardless of how the central claim resolves

Three results don't wait on an unconfounded joint-necessity test: QE-in-the-loop misses its threshold on 3/5 tested pairs, source-side triggering shows no signal on those same pairs, and BLOOMZ's generation reliability is gated by pretraining exposure (below, now five points). These are usable in the paper draft now, independent of which option we pick next.

Scope against the proposal

Three honesty notes. First, ha/km/ps/xh/zu were chosen because real WMT human judgments exist for them, not because they're confirmed as the Matrix Lab's actual target portfolio — worth checking those match before leaning on these results for portfolio-specific decisions. Second, these six phases only touch Secs. 5.1–5.3; document-level session management (Sec. 5.4), parallel hypothesis exploration (Sec. 5.5), and interactive interpretation with clarification (Sec. 5.6, the clinical MT program's home turf) are untested here. Third, Phase 1's own per-pair QE-reliability numbers mix translation directions (ha is en→ha; km/ps are X→en; xh/zu are never measured against English at all) — worth keeping in mind before citing any of them as evidence about the en→X direction this project actually tests.

Open decision

Three ways to get an unconfounded joint-necessity result:

1. **Keep searching** for an open model that is conversation-tuned and covers the target languages. Two candidates already ruled out; no known third, and preserves an all-open-source stack.

2. **Split the two roles**: an API model for reasoning, a Church-fine-tuned NLLB model queried as the tool. Sidesteps Phase 6's exposure floor (NLLB-200 targets all five pairs by design, though we have not verified its per-language training scale the way Phase 6 did for BLOOM/mT5 — that check should happen before committing to this option) and more faithfully tests the theory's actual claim (reasoning *and* tool use, not one model doing both). Costs API spend, drops the fully-open constraint.
3. **Narrow scope**: run the test on one pair BLOOMZ already generates reliably (en-fr, 100% clean), treat low-resource generalization as future work.
4. **Redesign Phase 3 around Aya-101's actual architecture, on Aya-101 — already hit a wall.** The original CoT confound wasn't just a bad prompt — decoder-only CoT assumes reasoning and source tokens share one self-attention stream, which doesn't hold for a T5 encoder-decoder (the decoder only sees the source through cross-attention to a representation the encoder already fixed). The redesign called for a mandatory *format-learning pre-test* before trusting any result built on structured output — ran it: 3 few-shot examples of "Analysis: ... / Translation: ...", 20 held-out en-ha sentences. $0/20$ **produced the requested structure.** Every response skipped straight to `Translation:`, ignoring the analysis step entirely despite three worked examples immediately before each prompt. This isn't a partial failure to work around — Aya-101 cannot currently be steered into intermediate reasoning content via few-shot prompting, so the 3-condition design this option describes is not executable on this model as specified. Would need either a different elicitation strategy (fine-tuning, forced decoding) or a different model — which folds this option back into Option 1's search, not a parallel path.

Leaning toward option 2 — it resolves Phase 6's confound directly rather than continuing to search for a model that clears it, and now more so: Option 4's pre-test result shows Aya-101 specifically can't support the redesigned unconfounded test either, so there's no cheap parallel path left on the current model. Option 3 is the fallback if preserving an open-only stack matters more than resolving the question now.

What changes in the paper draft

- **Sec. 5.3 (QE-in-the-loop)**: explicit caveat — doesn't work for 3/5 tested pairs; reframe as open problem.
- **Sec. 5.1 (triggering)**: same — cheap features show no signal; only crude implementations ruled out so far.
- **Sec. 1 (motivation)**: add the exposure-floor finding as a distinct failure mode from reasoning/tool insufficiency.
- **Core joint-necessity claim**, wherever stated: needs "confounded/untested cleanly" framing — not silence, not overclaiming from Phase 3.
- **Sec. 7 (evaluation protocol)**: the (b)-vs-(c) best-of-N comparison it specifies is now partially filled in for two pairs — en-ha (compute alone beat every architected condition, by 0.064) and en-de (same direction, but the gap shrinks to 0.007 under a more trustworthy QE judge). Best current read: en-ha's large gap is mostly QE noise, not a real compute-beats-architecture finding, but this is high-resource-vs-low-resource evidence, not a clean rerun of the same pair under two judges — no low-resource pair currently has validated en→X QE to run that clean version on.

Appendix: debugging record (compressed)

Every entry below is a bug that was caught only after it produced a confusing result. Phases 1 and 2's results have not had equivalent adversarial scrutiny applied to them — treat their clean appearance with that base rate in mind, not as proof

nothing is wrong.

- **Aya-101 (Phase 3)**: T5 seq2seq, no chat template; flattened multi-turn context to a role-tagged transcript — fine for single-turn conditions, degenerated for the 4-step CoT chain.
- **Refine-step scoring bug (Phase 4)**: Qwen2.5-32B's refine responses were often full English explanations, entirely scored as "the translation" — a fake collapse from $0.71 \rightarrow 0.53$ COMET-KIWI. Fixed with an "output only the translation" instruction plus an extractor, validated against the original buggy data ($287/290$ cleaned correctly).
- **BLOOMZ/InkubaLM (Phase 5)**: 9 fixes — missing pre-download, an outdated KV-cache path in InkubaLM's custom model class, partial gated-repo caching, a context-window miscalculation, role-tag framing that made both models echo prompt boilerplate, a cue-removal that caused empty output, a misdiagnosed repetition-penalty fix, a cue-induced truncation bug, and finally a genuine correct-vs.-echo reliability problem that wasn't further fixable. An earlier full run that looked like a contradiction of Phase 3 was explicitly retracted once these bugs were understood.
- **ROOTS-table correction (Phase 6)**: a secondary characterization claimed Xhosa/Zulu were absent from BLOOM's pretraining data. The primary source (BLOOM paper, Table 1) shows both present with real, small byte counts. The claim that German has *zero* presence held up, though — and turned out to be the more consequential finding.